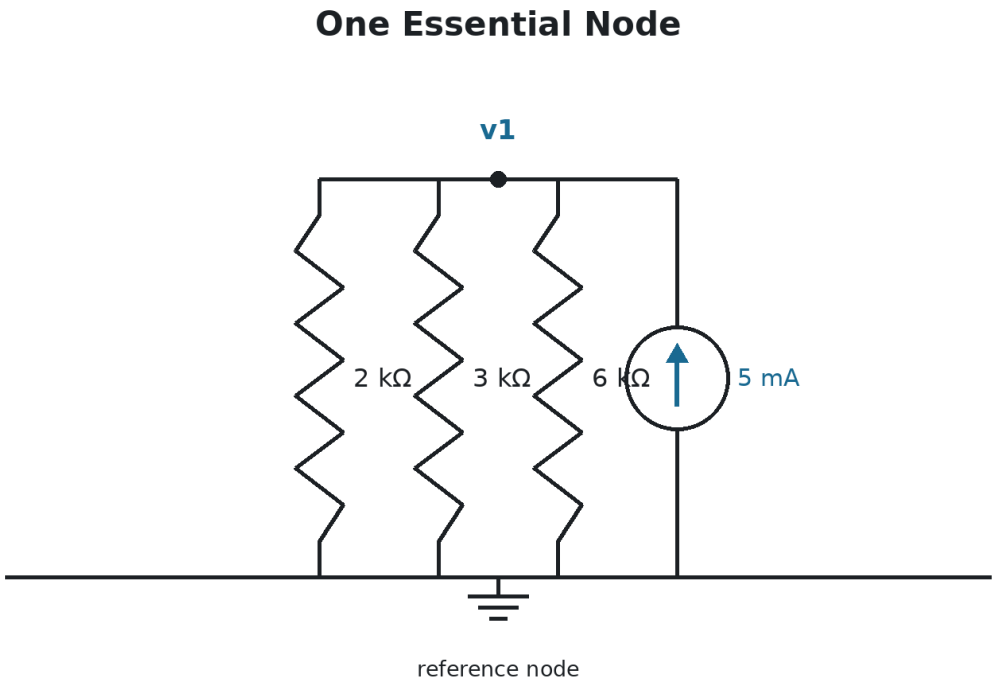
Nodal Questions

Question 1

A 5 mA source injects current upward into node 1. Three resistors of 2 kΩ, 3 kΩ, and 6 kΩ connect node 1 to ground. Find the node voltage and state the current direction in the 3 kΩ resistor.



Node 1 has parallel 2 kΩ, 3 kΩ, and 6 kΩ branches to ground and a 5 mA source pointing upward from ground.

Solution

Let the labeled voltages be measured relative to ground. Take currents leaving each node as positive. Use kΩ and mA, so the computed voltages are in volts.

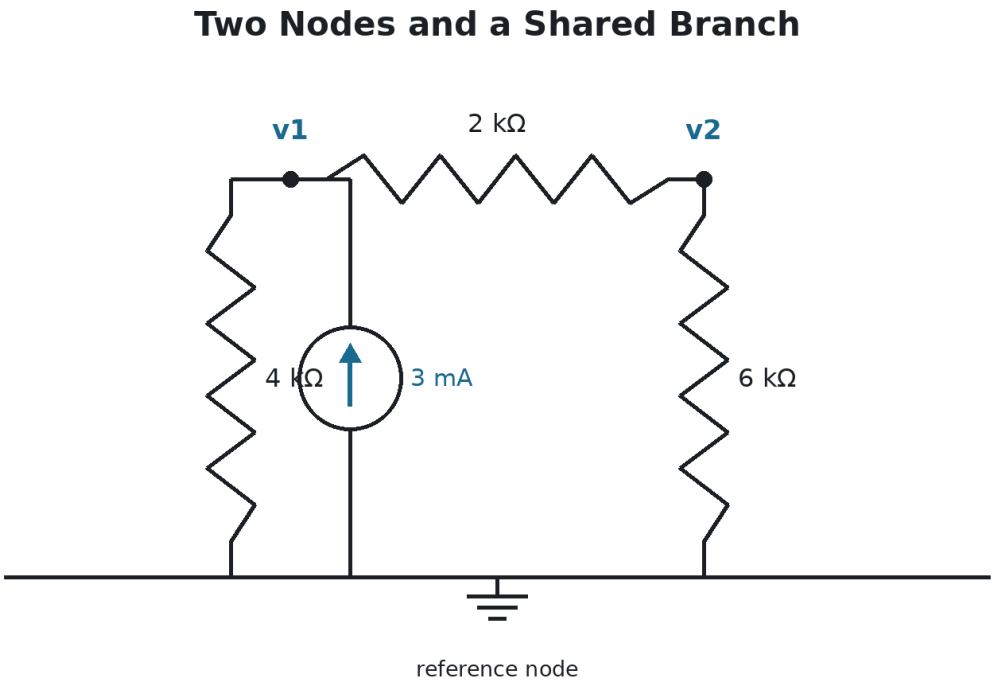
At node 1: $v_1/2 + v_1/3 + v_1/6 - 5 = 0$

Solving the simultaneous equations gives $v_1 = 5.000$ V $i_{3\text{k}} = v_1/(3\ \text{k}\Omega) = 1.667$ mA

The positive result means the current flows downward from node 1 to ground.

Question 2

A 3 mA source injects current into node 1. Node 1 has a 4 kΩ branch to ground and a 2 kΩ branch to node 2. Node 2 has a 6 kΩ branch to ground. Find both node voltages and the current from node 1 to node 2.



A 3 mA source points from ground to node 1; 4 kΩ joins node 1 to ground, 2 kΩ joins the nodes, and 6 kΩ joins node 2 to ground.

Solution

Let the labeled voltages be measured relative to ground. Take currents leaving each node as positive. Use kΩ and mA, so the computed voltages are in volts.

At node 1: $v_1/4 + (v_1 - v_2)/2 - 3 = 0$

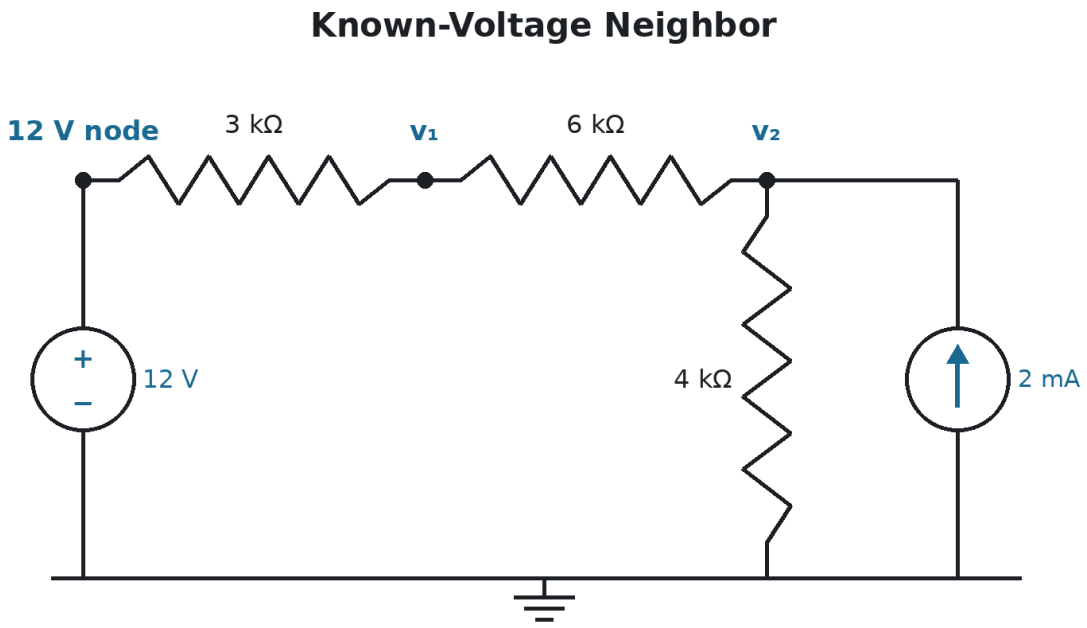
At node 2: $v_2/6 + (v_2 - v_1)/2 = 0$

Solving the simultaneous equations gives $v_1 = 8.000$ V, $v_2 = 6.000$ V $i_{12} = (v_1 - v_2)/(2\ \text{k}\Omega) = 1.000$ mA

The current flows from node 1 to node 2.

Question 3

A 3 kΩ resistor connects a fixed 12 V node to node 1. A 6 kΩ resistor joins node 1 to node 2. Node 2 connects to ground through 4 kΩ, and a 2 mA source injects current from ground into node 2. Find both unknown node voltages.



A fixed 12 V node connects through 3 kΩ to node 1; 6 kΩ joins nodes 1 and 2; node 2 has a 4 kΩ branch and an upward 2 mA source to ground.

Solution

Let the labeled voltages be measured relative to ground. Take currents leaving each node as positive. Use kΩ and mA, so the computed voltages are in volts.

At node 1: $(v_1 - 12)/3 + (v_1 - v_2)/6 = 0$

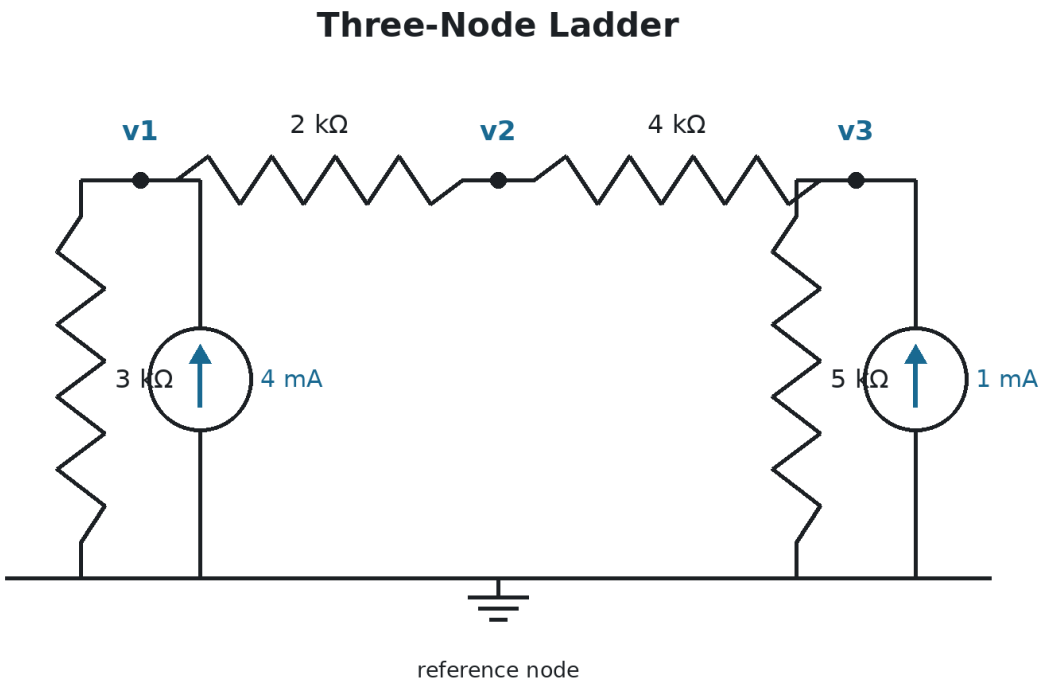
At node 2: $(v_2 - v_1)/6 + v_2/4 - 2 = 0$

Solving the simultaneous equations gives $v_1 = 11.077$ V, $v_2 = 9.231$ V $i_{12} = (v_1 - v_2)/(6\ \text{k}\Omega) = 0.308$ mA

The current flows from node 1 to node 2.

Question 4

Three nodes form a ladder: 2 kΩ joins nodes 1 and 2, and 4 kΩ joins nodes 2 and 3. Node 1 has 3 kΩ to ground and receives 4 mA. Node 3 has 5 kΩ to ground and receives 1 mA. Find all node voltages.



Node 1 has 3 kΩ to ground and a 4 mA upward source; 2 kΩ joins nodes 1–2, 4 kΩ joins nodes 2–3, and node 3 has 5 kΩ plus a 1 mA upward source to ground.

Solution

Let the labeled voltages be measured relative to ground. Take currents leaving each node as positive. Use kΩ and mA, so the computed voltages are in volts.

At node 1: $v_1/3 + (v_1 - v_2)/2 - 4 = 0$

At node 2: $(v_2 - v_1)/2 + (v_2 - v_3)/4 = 0$

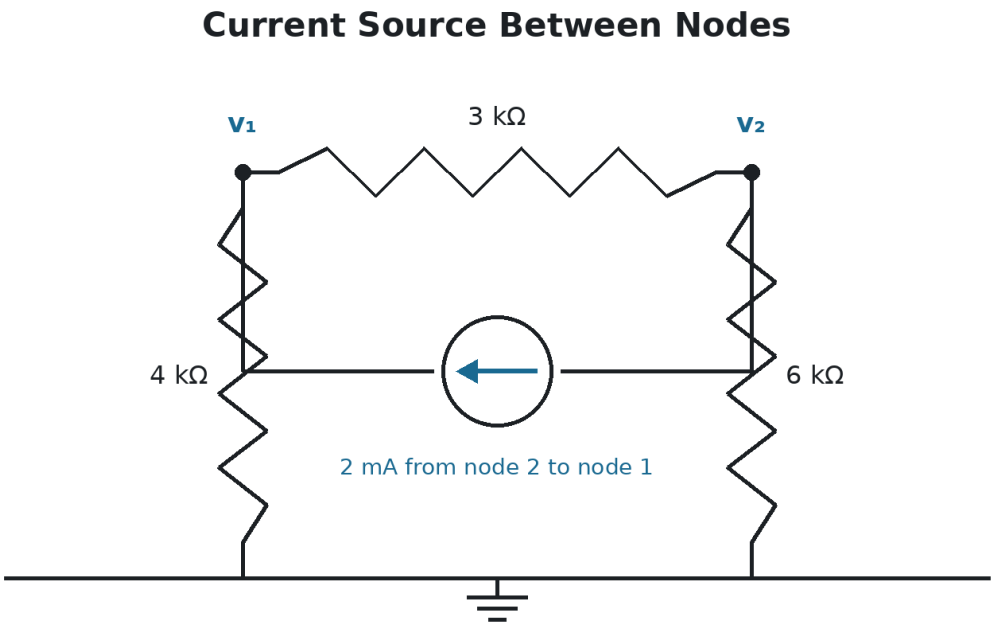
At node 3: $v_3/5 + (v_3 - v_2)/4 - 1 = 0$

Solving the simultaneous equations gives $v_1 = 10.500$ V, $v_2 = 9.500$ V, $v_3 = 7.500$ V $i_{23} = (v_2 - v_3)/(4\ \text{k}\Omega) = 0.500$ mA

The current flows from node 2 to node 3.

Question 5

Nodes 1 and 2 are joined by a 3 kΩ resistor and a 2 mA source directed from node 2 to node 1. Node 1 has 4 kΩ to ground; node 2 has 6 kΩ to ground. Find both node voltages and interpret their signs.



A 3 kΩ resistor and a 2 mA source directed from node 2 toward node 1 lie between the nodes; 4 kΩ joins node 1 to ground and 6 kΩ joins node 2 to ground.

Solution

Let the labeled voltages be measured relative to ground. Take currents leaving each node as positive. Use kΩ and mA, so the computed voltages are in volts.

At node 1: $v_1/4 + (v_1 - v_2)/3 - 2 = 0$

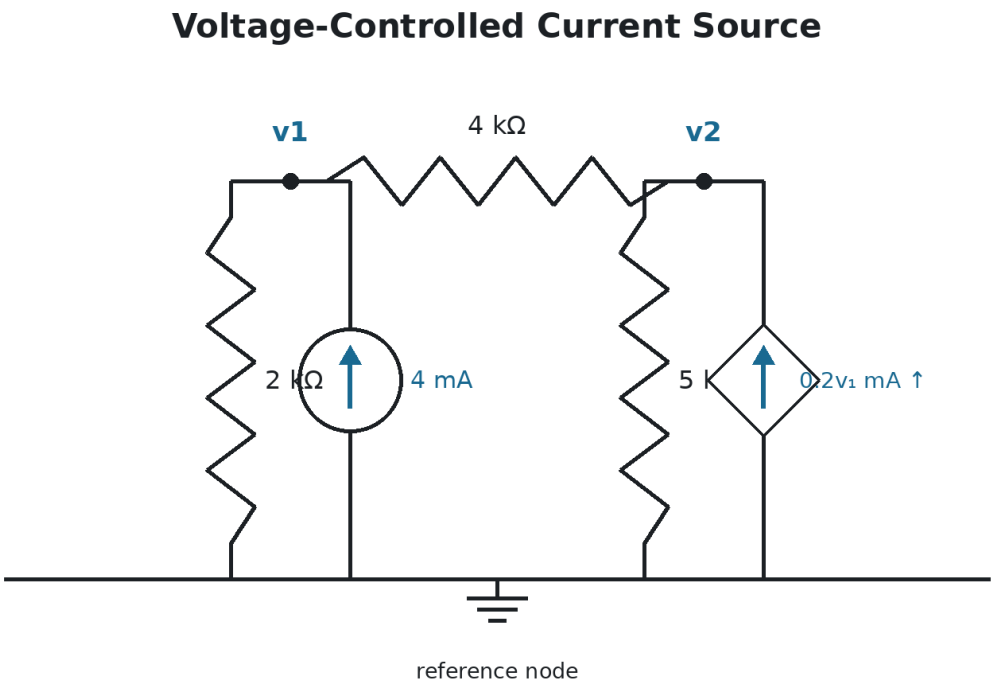
At node 2: $v_2/6 + (v_2 - v_1)/3 + 2 = 0$

Solving the simultaneous equations gives $v_1 = 1.846$ V, $v_2 = -2.769$ V $i_{4\text{k}} = v_1/(4\ \text{k}\Omega) = 0.462$ mA

Node 2 is negative because its voltage is below the chosen reference. Its 6 kΩ branch current actually flows upward.

Question 6

A 4 mA source injects node 1. Node 1 has 2 kΩ to ground and 4 kΩ to node 2. Node 2 has 5 kΩ to ground and an upward VCCS of 0.2 times node voltage 1 in mA. Find both voltages and the controlled current.



Node 1 receives 4 mA and has 2 kΩ to ground; 4 kΩ joins the nodes; node 2 has 5 kΩ to ground and an upward VCCS equal to 0.2 times node voltage 1 in mA.

Solution

Let the labeled voltages be measured relative to ground. Take currents leaving each node as positive. Use kΩ and mA, so the computed voltages are in volts.

At node 1: $v_1/2 + (v_1 - v_2)/4 - 4 = 0$

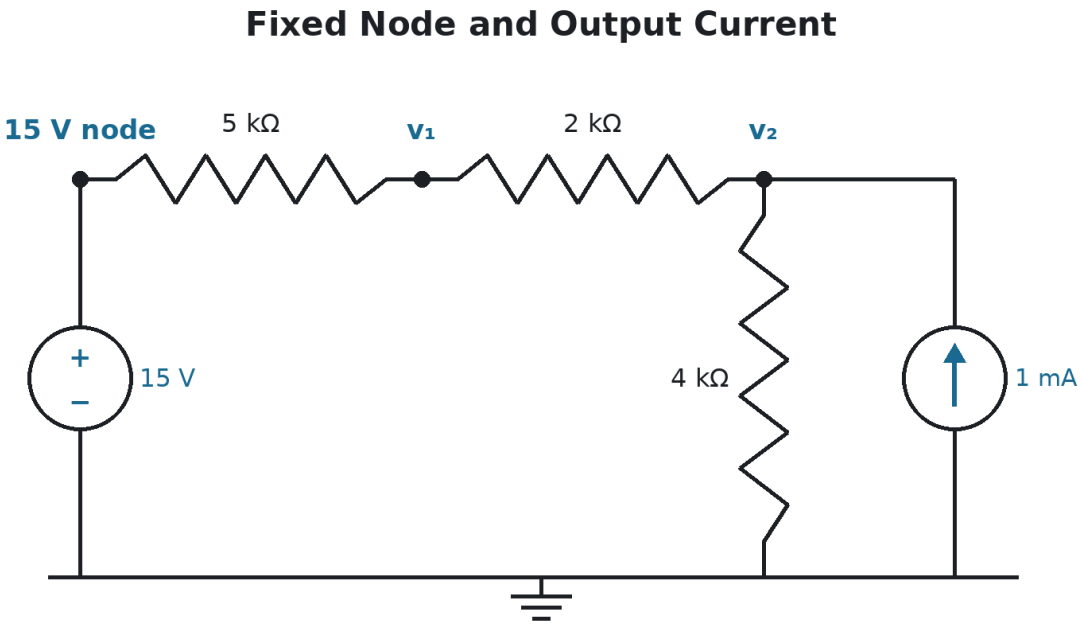
At node 2: $v_2/5 + (v_2 - v_1)/4 - 0.2v_1 = 0$

Solving the simultaneous equations gives $v_1 = 8.000$ V, $v_2 = 8.000$ V $i_d = (0.2\ \text{mS})v_1 = 1.600$ mA

The controlled current points upward as drawn. Because the node voltages are equal, the 4 kΩ branch current is zero.

Question 7

A 5 kΩ resistor connects a 15 V node to node 1. A 2 kΩ resistor joins node 1 to node 2. Node 2 has a 4 kΩ resistor to ground and receives 1 mA from ground. Find both voltages and the current drawn from the 15 V node.



The fixed 15 V node connects through 5 kΩ to node 1; 2 kΩ joins the unknown nodes; node 2 has 4 kΩ and a 1 mA upward source to ground.

Solution

Let the labeled voltages be measured relative to ground. Take currents leaving each node as positive. Use kΩ and mA, so the computed voltages are in volts.

At node 1: $(v_1 - 15)/5 + (v_1 - v_2)/2 = 0$

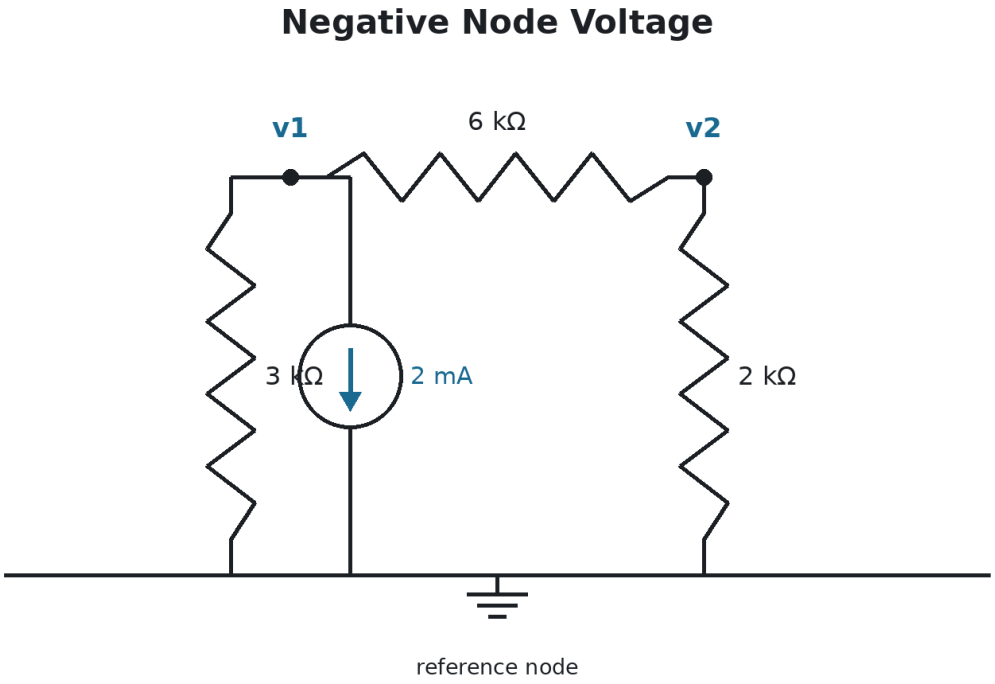
At node 2: $(v_2 - v_1)/2 + v_2/4 - 1 = 0$

Solving the simultaneous equations gives $v_1 = 10.000$ V, $v_2 = 8.000$ V $i_{15} = (15 - v_1)/(5\ \text{k}\square) = 1.000$ mA

The 15 V node supplies 1.000 mA toward node 1.

Question 8

A 2 mA source points downward from node 1 to ground. Node 1 has 3 kΩ to ground and 6 kΩ to node 2. Node 2 has 2 kΩ to ground. Find both voltages and explain the negative sign.



A downward 2 mA source and 3 kΩ resistor join node 1 to ground; 6 kΩ joins the nodes; 2 kΩ joins node 2 to ground.

Solution

Let the labeled voltages be measured relative to ground. Take currents leaving each node as positive. Use kΩ and mA, so the computed voltages are in volts.

At node 1: $v_1/3 + (v_1 - v_2)/6 + 2 = 0$

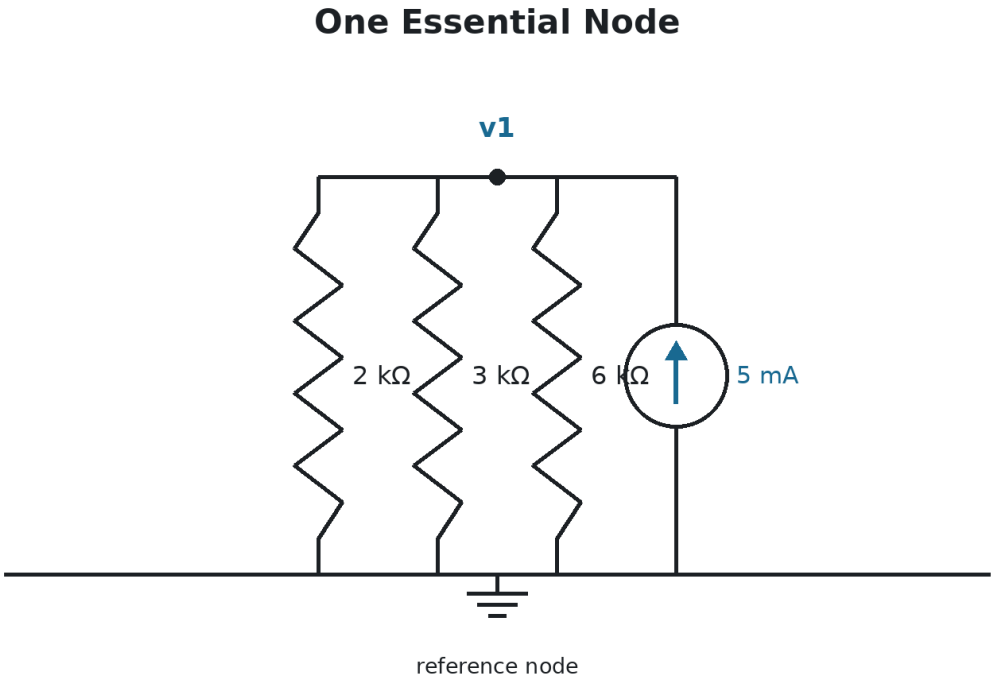
At node 2: $v_2/2 + (v_2 - v_1)/6 = 0$

Solving the simultaneous equations gives $v_1 = -4.364$ V, $v_2 = -1.091$ V $i_{12} = (v_1 - v_2)/(6\ \text{k}\Omega) = -0.545$ mA

Relative to the stated node-1-to-node-2 direction, the result is negative. The actual current is 0.545 mA from node 2 to node 1.

Question 9

Write the nodal equation for the one-node circuit, solve for the voltage, and verify that the resistor currents sum to the 5 mA source current.



Node 1 has parallel 2 kΩ, 3 kΩ, and 6 kΩ branches to ground and a 5 mA source pointing upward from ground.

Solution

Let v_1 be relative to ground and take currents leaving as positive.

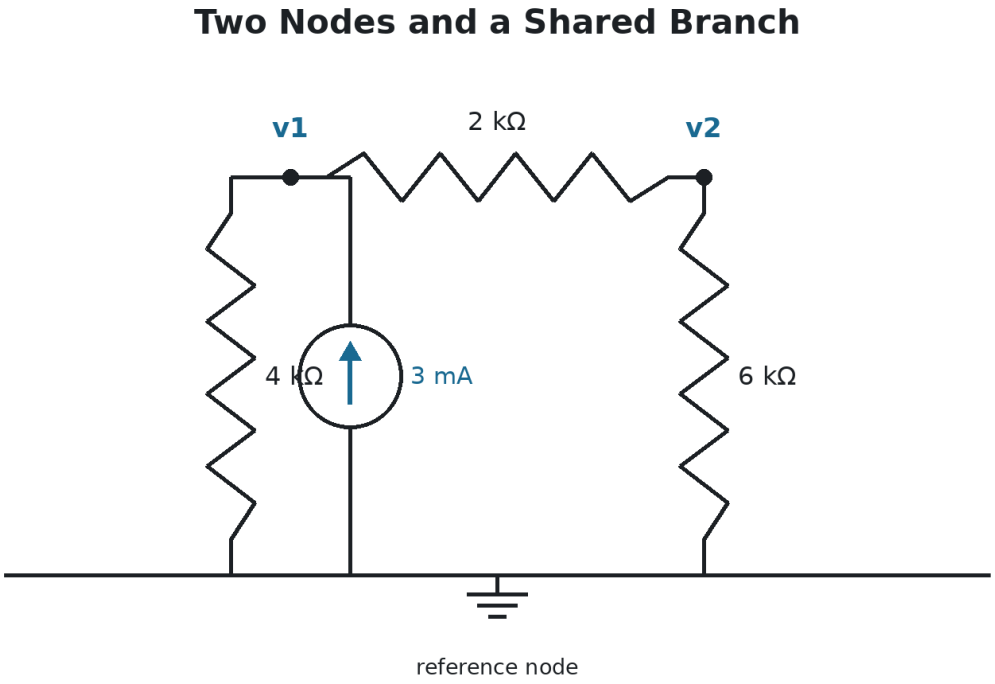
$\frac{v_1}{2} + \frac{v_1}{3} + \frac{v_1}{6} - 5 = 0$ $v_1 = 5.000$ V

The three downward currents are $i_2 = 2.500$ mA, $i_3 = 1.667$ mA, $i_6 = 0.833$ mA $i_2 + i_3 + i_6 = 5.000$ mA

KCL is satisfied.

Question 10

For the two-node circuit, form the conductance matrix and solve for the voltage across the 2 kΩ resistor, defined positive on the node 1 side.



A 3 mA source points from ground to node 1; 4 kΩ joins node 1 to ground, 2 kΩ joins the nodes, and 6 kΩ joins node 2 to ground.

Solution

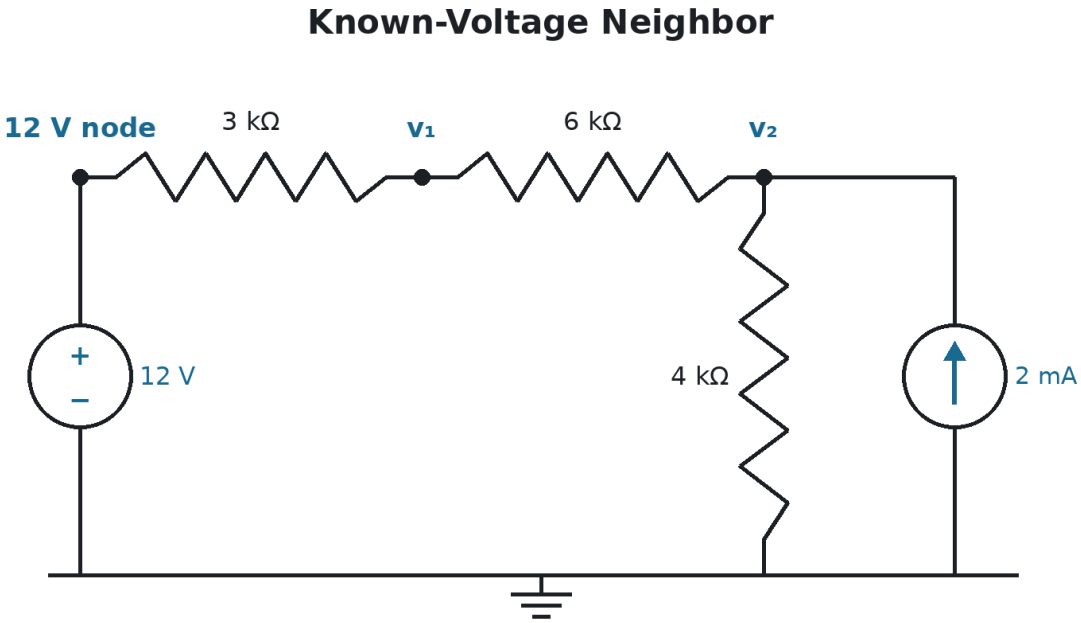
Take currents leaving each node as positive. In mS, the matrix equation is $\begin{bmatrix} 0.75 & -0.50 \\ -0.50 & 0.667 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 3 \\ 0 \end{bmatrix}$

Solving gives $v_1 = 8.000$ V, $v_2 = 6.000$ V

With the stated polarity, $v_{12} = v_1 - v_2 = 2.000$ V

Question 11

Find the current and power of the 12 V source. State whether it absorbs or delivers power.



A 12 V source fixes the left node; 3 kΩ joins it to node 1, 6 kΩ joins nodes 1–2, and node 2 has 4 kΩ plus an upward 2 mA source to ground.

Solution

Take currents leaving each unknown node as positive.

$\frac{v_1-12}{3}+\frac{v_1-v_2}{6}=0$

$\frac{v_2-v_1}{6}+\frac{v_2}{4}-2=0$ $v_1=11.077$ V, $v_2=9.231$ V

Current leaves the positive terminal of the 12 V source:

$i_s=\frac{12-v_1}{3\ \text{k}\Omega}=0.308$ mA

Using the passive sign convention,

$p_{12}=-\left(12\ \mathrm{V}

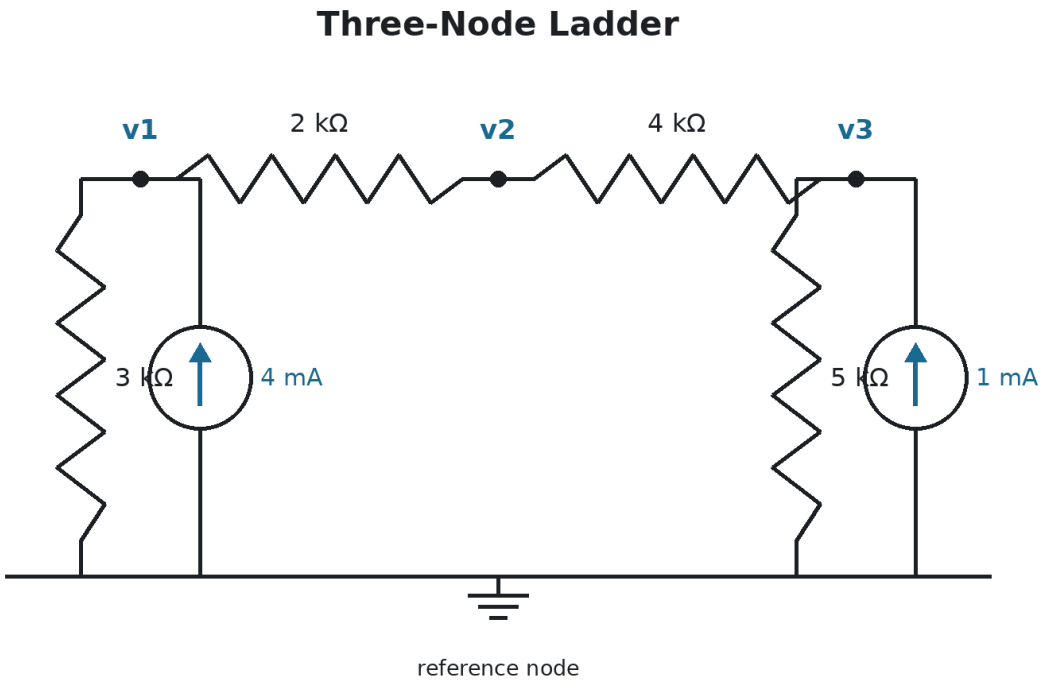
ight)\left(0.308\ \mathrm{mA}

ight)=-3.69\ \mathrm{mW}$

The source delivers 3.69 mW.

Question 12

Use nodal analysis to find the voltage and current of the middle 4 kΩ branch. Define current from node 2 toward node 3.



Node 1 has 3 kΩ and an upward 4 mA source to ground; 2 kΩ joins nodes 1–2, 4 kΩ joins nodes 2–3, and node 3 has 5 kΩ plus an upward 1 mA source.

Solution

Take currents leaving each node as positive.

$\frac{v_1}{3} + \frac{v_1 - v_2}{2} - 4 = 0$

$\frac{v_2 - v_1}{2} + \frac{v_2 - v_3}{4} = 0$

$\frac{v_3}{5} + \frac{v_3 - v_2}{4} - 1 = 0$ $v_1 = 10.500$ V, $v_2 = 9.500$ V, $v_3 = 7.500$ V $v_{23} = v_2 - v_3 = 2.000$ V

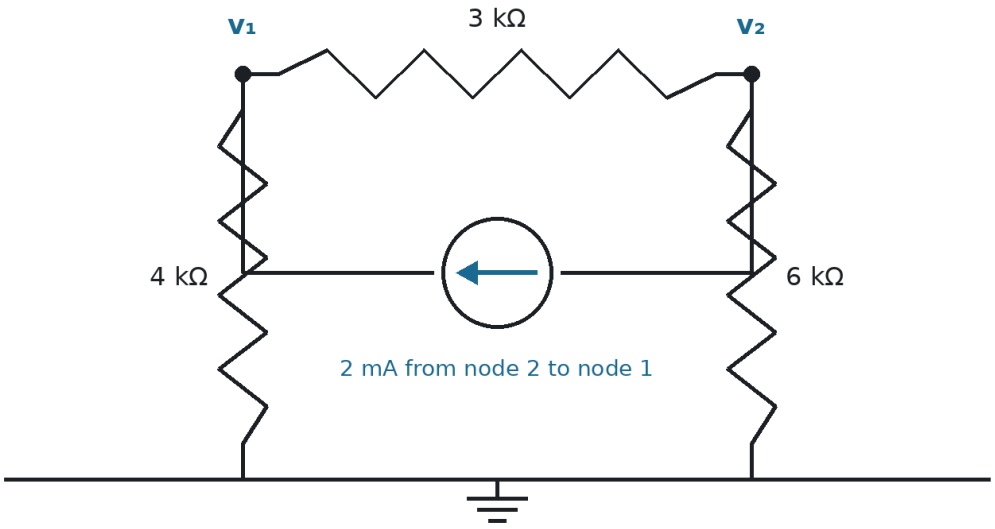
$i_{23} = \frac{v_{23}}{4\ \text{k}\Omega} = 0.500$ mA

The current flows from node 2 to node 3.

Question 13

Determine the power of the 2 mA source between the nodes. Use the passive sign convention and state absorbed or delivered.

Current Source Between Nodes



A 3 kΩ resistor and a 2 mA source directed from node 2 to node 1 lie between the nodes; 4 kΩ joins node 1 to ground and 6 kΩ joins node 2 to ground.

Solution

Take currents leaving each node as positive.

$\frac{v_1}{4} + \frac{v_1 - v_2}{3} - 2 = 0$

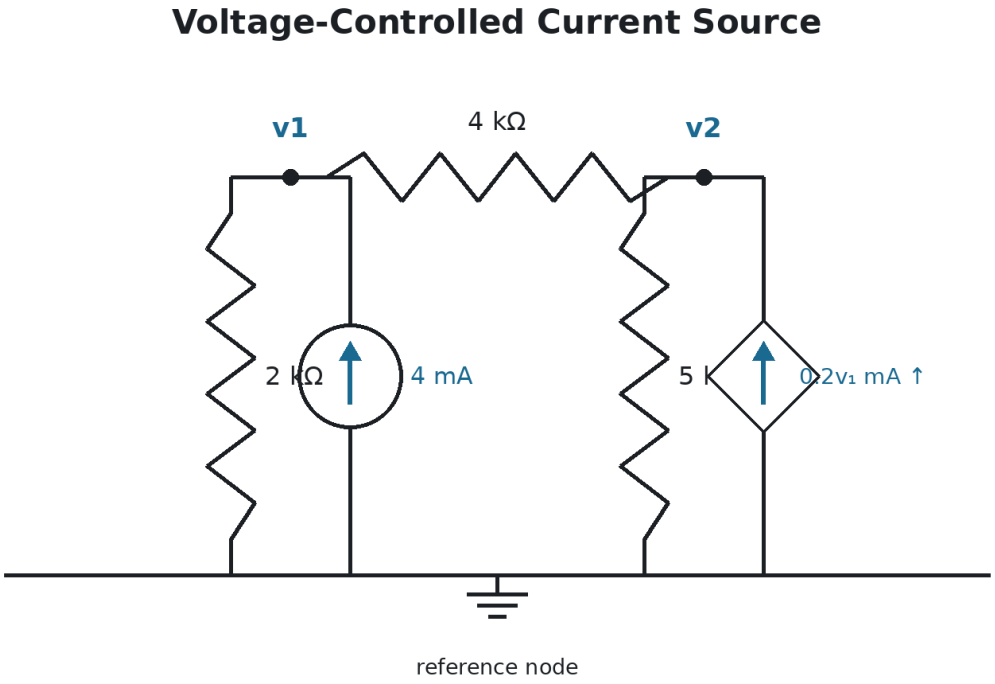
$\frac{v_2}{6} + \frac{v_2 - v_1}{3} + 2 = 0$ $v_1 = 1.846$ V, $v_2 = -2.769$ V

The source current enters its node 2 terminal. Its passive voltage is $v_s = v_2 - v_1 = -4.615$ V $p_s = v_s(2\text{ mA}) = -9.23$ mW

The source delivers 9.23 mW.

Question 14

Find the power of the dependent current source. State whether it absorbs or delivers power.



Node 1 receives 4 mA and has 2 kΩ to ground; 4 kΩ joins the nodes; node 2 has 5 kΩ and an upward VCCS equal to 0.2 times node voltage 1 in mA.

Solution

Take currents leaving each node as positive.

$\frac{v_1}{2} + \frac{v_1 - v_2}{4} - 4 = 0$

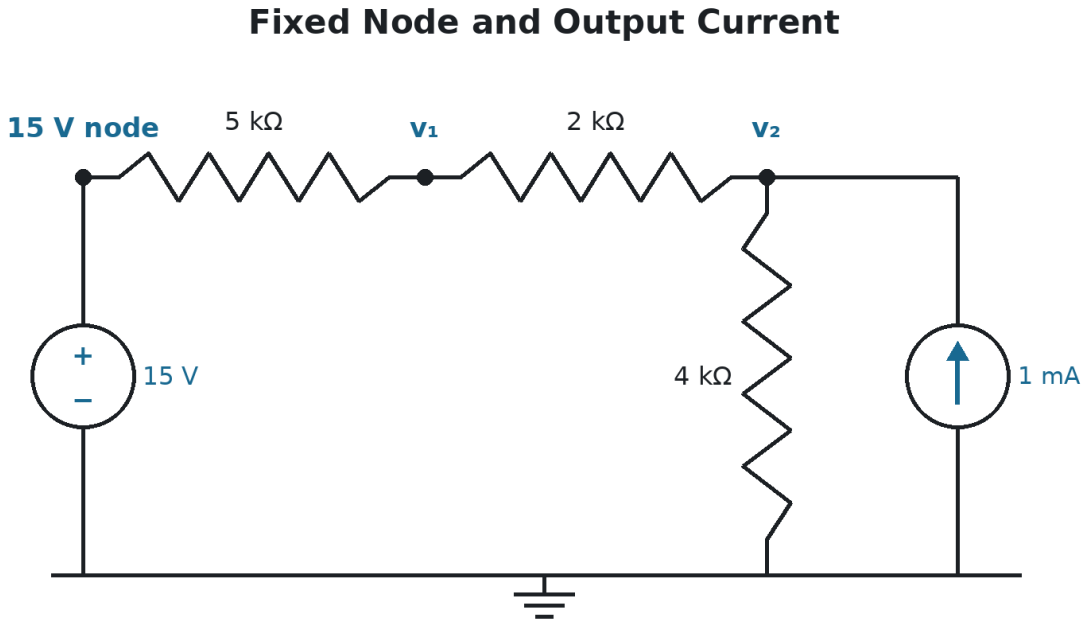
$\frac{v_2}{5} + \frac{v_2 - v_1}{4} - 0.2v_1 = 0$ $v_1 = v_2 = 8.000$ V $i_d = (0.2\ \text{mS})v_1 = 1.600$ mA

The upward current enters the source at its ground terminal, so the passive voltage is $-v_2$. $p_d = (-8.000\ \text{V})(1.600\ \text{mA}) = -12.8$ mW

The dependent source delivers 12.8 mW.

Question 15

Find the power delivered by the 15 V source and the power absorbed by the 4 kΩ resistor.



The 15 V source fixes the left node; 5 kΩ joins it to node 1, 2 kΩ joins the unknown nodes, and node 2 has 4 kΩ plus an upward 1 mA source.

Solution

Take currents leaving each unknown node as positive.

$\frac{v_1-15}{5} + \frac{v_1-v_2}{2} = 0$

$\frac{v_2-v_1}{2} + \frac{v_2}{4} - 1 = 0$ $v_1 = 10.000$ V, $v_2 = 8.000$ V

$i_{15} = \frac{15-v_1}{5\ \text{k}\Omega} = 1.000$ mA

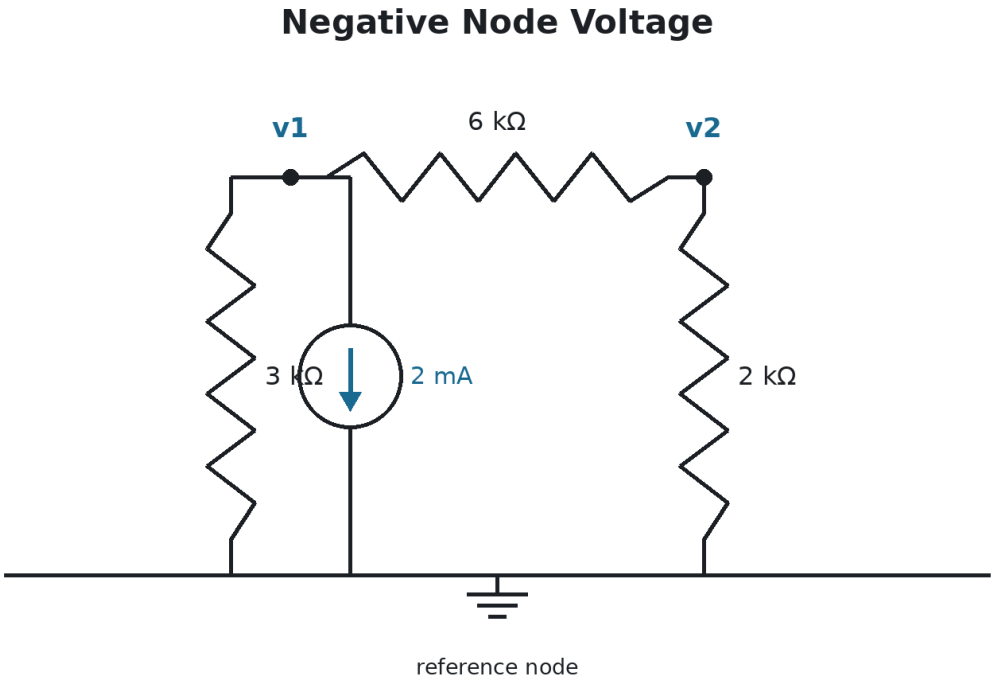
Current leaves the positive terminal, so $p_{15} = -(15\ \text{V})(1.000\ \text{mA}) = -15.0$ mW

The source delivers 15.0 mW. The resistor absorbs

$p_{4\text{k}} = \frac{v_2^2}{4\ \text{k}\Omega} = 16.0$ mW

Question 16

The reference node is unchanged. Find the voltage across the 6 kΩ resistor with positive polarity at node 2, then state the actual current direction.



A downward 2 mA source and 3 kΩ resistor join node 1 to ground; 6 kΩ joins the nodes; 2 kΩ joins node 2 to ground.

Solution

Take currents leaving each node as positive.

$\frac{v_1}{3} + \frac{v_1 - v_2}{6} + 2 = 0$

$\frac{v_2}{2} + \frac{v_2 - v_1}{6} = 0$ $v_1 = -4.364$ V, $v_2 = -1.091$ V

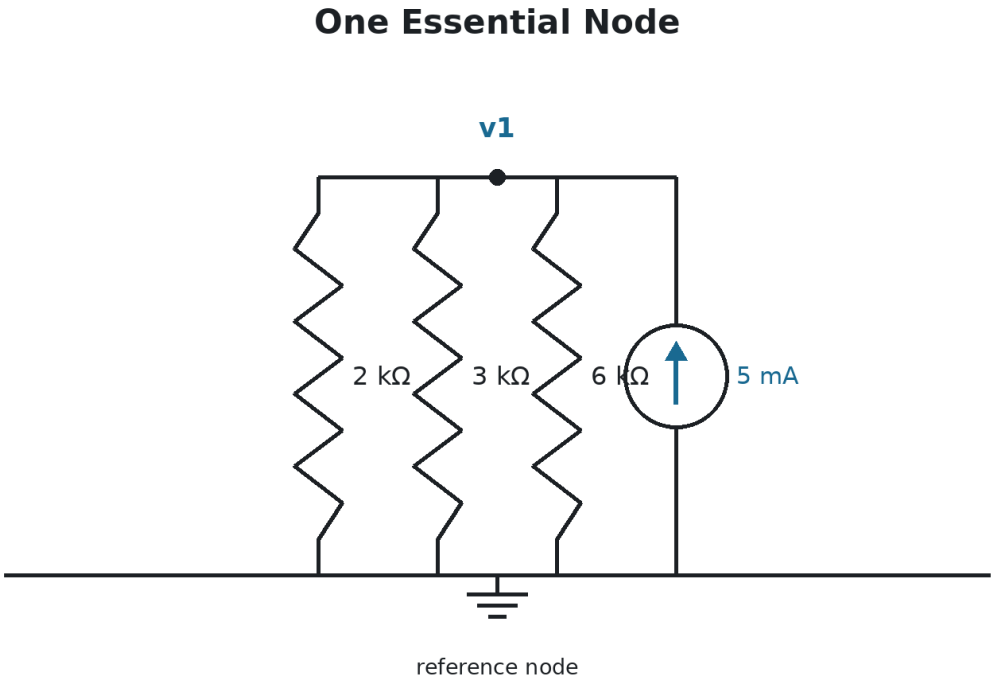
With positive polarity at node 2, $v_{21} = v_2 - v_1 = 3.273$ V

$i_{21} = \frac{v_{21}}{6\ \text{k}\Omega} = 0.545$ mA

The actual current flows from node 2 to node 1.

Question 17

For the one-node circuit, find the equivalent resistance seen from node 1 and use it to compute the node voltage.



Node 1 has parallel 2 kΩ, 3 kΩ, and 6 kΩ branches to ground and a 5 mA source pointing upward from ground.

Solution

The three resistors are in parallel from node 1 to ground.

$R_{eq}=\left(\frac{1}{2}+\frac{1}{3}+\frac{1}{6}

ight)^{-1}\mathrm{k\Omega}=1.000\ \mathrm{k\Omega}$

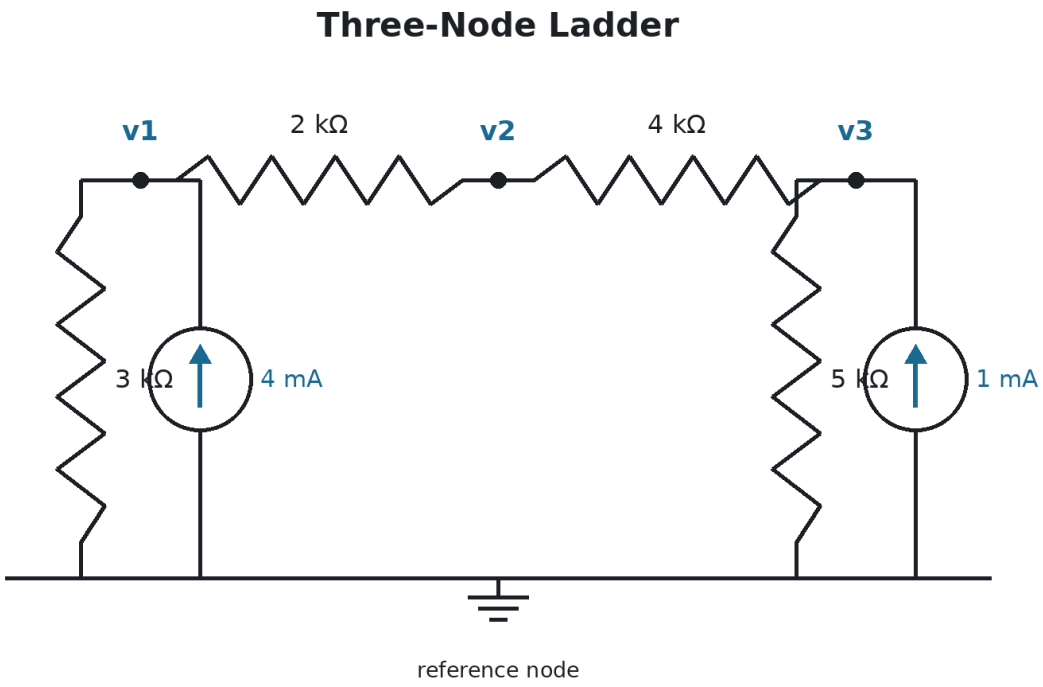
The 5 mA source drives this equivalent resistance: $v_1 = (5.000$ mA$)(1.000$ k$\square) = 5.000$ V

This agrees with direct KCL:

$\frac{v_1}{2} + \frac{v_1}{3} + \frac{v_1}{6} = 5$

Question 18

Find the total power absorbed by all four resistors in the ladder and the power delivered by both current sources. Verify power balance.



Node 1 has 3 kΩ and an upward 4 mA source to ground; 2 kΩ joins nodes 1–2, 4 kΩ joins nodes 2–3, and node 3 has 5 kΩ plus an upward 1 mA source.

Solution

Using currents leaving each node as positive gives $v_1 = 10.500$ V, $v_2 = 9.500$ V, $v_3 = 7.500$ V

The resistor powers are

$p_{3\text{k}} = \frac{v_1^2}{3\ \text{k}\Omega} = 36.75$ mW

$p_{2\text{k}} = \frac{(v_1 - v_2)^2}{2\ \text{k}\Omega} = 0.50$ mW

$p_{4\text{k}} = \frac{(v_2 - v_3)^2}{4\ \text{k}\Omega} = 1.00$ mW

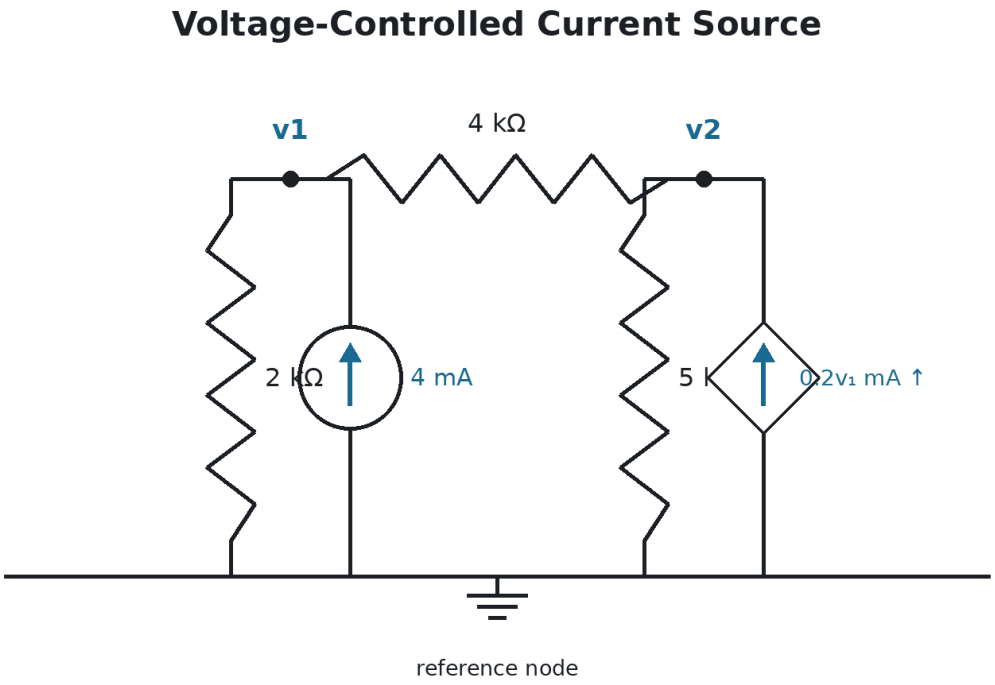
$p_{5\text{k}} = \frac{v_3^2}{5\ \text{k}\Omega} = 11.25$ mW $P_R = 49.50$ mW

Both source currents enter their negative-voltage terminals: $p_{4\text{mA}} = -v_1(4\ \text{mA}) = -42.0$ mW $p_{1\text{mA}} = -v_3(1\ \text{mA}) = -7.50$ mW

Thus the sources deliver 49.50 mW, equal to the absorbed power.

Question 19

Suppose the controlled-source coefficient is written as an unknown transconductance. Find the transconductance that makes the current in the 4 kΩ resistor zero.



Node 1 receives 4 mA and has 2 kΩ to ground; 4 kΩ joins the nodes; node 2 has 5 kΩ and an upward VCCS equal to an unknown transconductance times node voltage 1.

Solution

Zero current in the 4 kΩ resistor requires $v_1 = v_2 = v$

At node 1:

$\frac{v}{2} - 4 = 0 \; v = 8.000$ V

At node 2, with the controlled current directed upward:

$\frac{v}{5} - g_m v = 0$

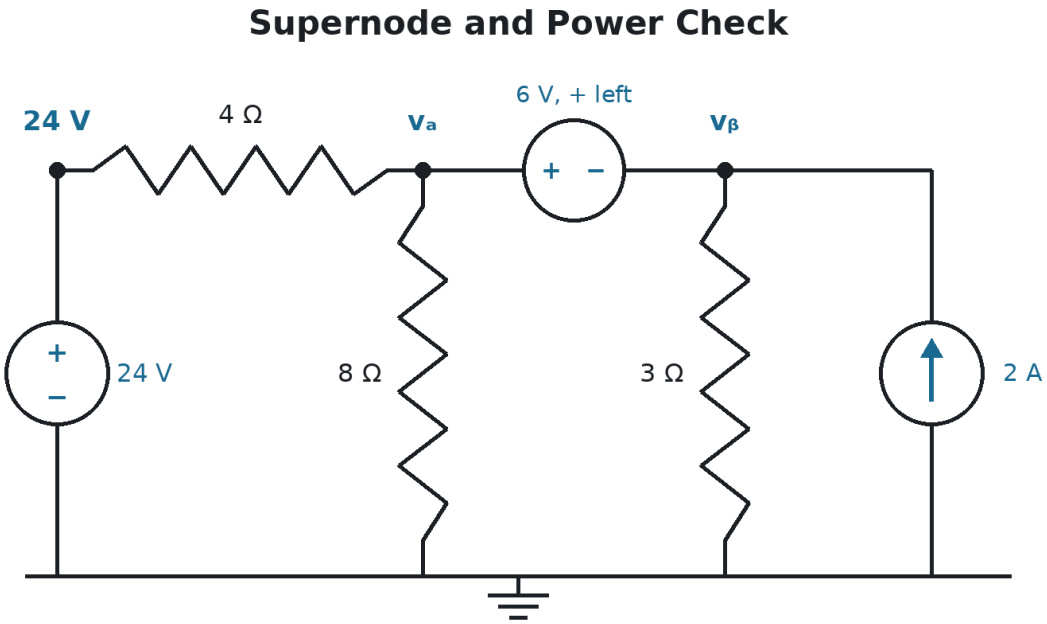
For nonzero voltage,

$g_m = \frac{1}{5\ \text{k}\Omega} = 0.200$ mS

The required transconductance is 0.200 mS.

Question 20

A 24 V source fixes the left node. A 4 Ω resistor connects it to node a. An ideal 6 V source lies between node a, its positive terminal, and node b. Node a has 8 Ω to ground. Node b has 3 Ω to ground and receives 2 A upward from ground. Use a supernode to find both node voltages, the 6 V source power, and a complete power check.



The bottom wire is ground. A 24 V source fixes the left node; 4 Ω joins it to node a; a 6 V source with positive terminal at node a joins node a to node b; 8 Ω joins node a to ground; 3 Ω and an upward 2 A source join node b to ground.

Solution

Nodes v_a and v_b form a supernode because an ideal 6 V source lies between them. Take currents leaving the supernode as positive.

$\frac{v_a-24}{4} + \frac{v_a}{8} + \frac{v_b}{3} - 2 = 0$

The source constraint is $v_a - v_b = 6$

Solving gives $v_a = 14.118$ V, $v_b = 8.118$ V

The current through the 6 V source, defined from v_a to v_b, follows KCL at v_a:

$i_6 = \frac{24-v_a}{4} - \frac{v_a}{8} = 0.706$ A

It enters the positive terminal, so $p_6 = (6\text{ V})(0.706\text{ A}) = 4.24$ W

The 6 V source absorbs power. For a full check: $p_{4\Omega} = 24.42$ W, $p_{8\Omega} = 24.91$ W, $p_{3\Omega} = 21.97$ W $p_{24} = -59.29$ W, $p_{2\text{A}} = -16.24$ W $\sum p = 24.42 + 24.91 + 21.97 + 4.24 - 59.29 - 16.24 = 0.01$ W ≈ 0

The small difference is rounding; the 24 V and 2 A sources deliver power.